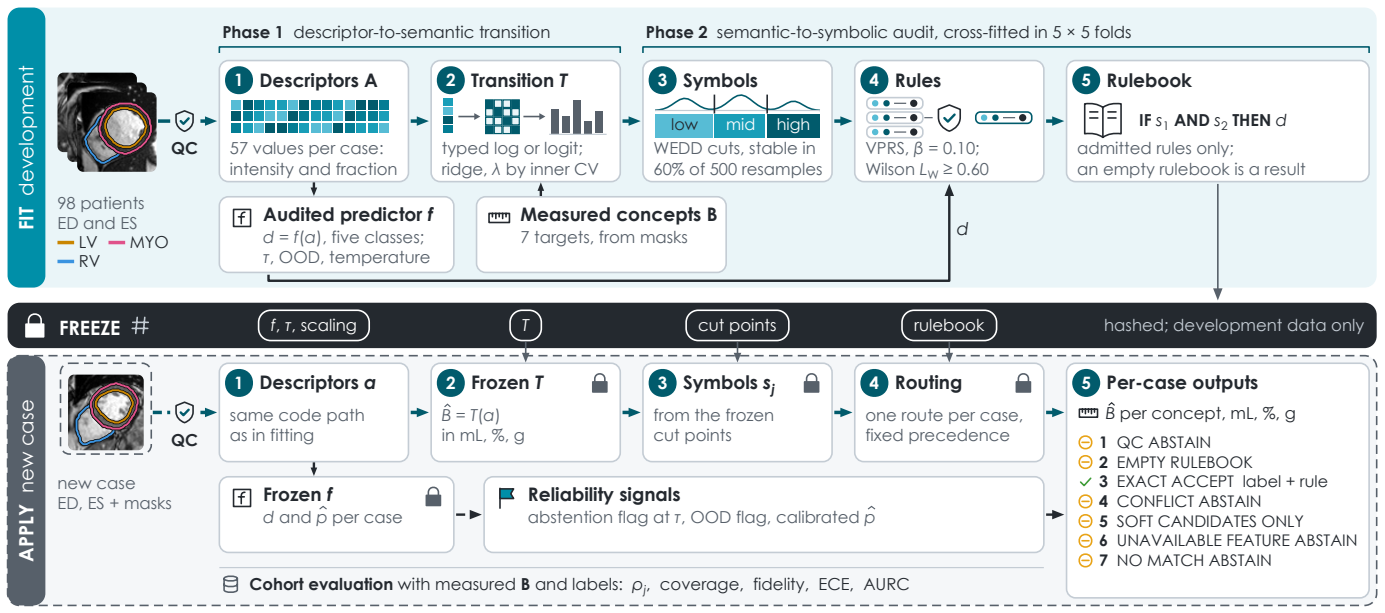

FIT development
Phase 1 descriptor-to-semantic transition
Phase 2 semantic-to-symbolic audit, cross-fitted in 5 × 5 folds
98 patients
ED and ES
LV
MYO
RV
QC
1 Descriptors A
57 values per case:
intensity and fraction
2 Transition T
typed log or logit;
ridge, λ by inner CV
3 Symbols
low
mid
high
WEDD cuts, stable in
60% of 500 resamples
4 Rules
VPRS, β = 0.10;
Wilson L_W ≥ 0.60
5 Rulebook
IF s1 AND s2 THEN d
admitted rules only;
an empty rulebook is a result
Audited predictor f
d = f(a), five classes;
τ, OOD, temperature
Measured concepts B
7 targets, from masks
d
FREEZE #
f, τ, scaling
T
cut points
rulebook
hashed; development data only
APPLY new case
new case
ED, ES + masks
QC
1 Descriptors a
same code path
as in fitting
2 Frozen T
B̂ = T(a)
in mL, %, g
3 Symbols s_j
from the frozen
cut points
4 Routing
one route per case,
fixed precedence
5 Per-case outputs
B̂ per concept, mL, %, g
1 QC ABSTAIN
2 EMPTY RULEBOOK
3 EXACT ACCEPT label + rule
4 CONFLICT ABSTAIN
5 SOFT CANDIDATES ONLY
6 UNAVAILABLE FEATURE ABSTAIN
7 NO MATCH ABSTAIN
Frozen f
d and p̂ per case
Reliability signals
abstention flag at τ, OOD flag, calibrated p̂
Cohort evaluation with measured B and labels: ρ_j, coverage, fidelity, ECE, AURC